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# Bihar Engineering University, Patna

## B.Tech. 1st Semester Examination, 2024 (Old)

***Code: 101102*** | ***Subject: Mathematics - I (Calculus, Multivariable Calculus & Linear Algebra)*** | ***Time: 03 Hours*** | ***Full Marks: 70***

***Instructions:-***

*(i)* *The marks are indicated in the right-hand margin.*
*(ii)* *There are NINE questions in this paper.*
*(iii)* *Attempt FIVE questions in all.*
*(iv)* *Question No. 1 is compulsory.*

***Q.1 Answer the following questions (Any seven only):*** **[14]**

(a) Show that $f(x)=\sin^2 x$ is continuous for every value of x.

(b) Verify Rolle's Theorem for the function $f(x)=2+(x-1)^{2/3}$ where $x$in[0,2].

(c) Test the convergence of the integral $\int_0^{\infty}[\cos x] / [1+x^2]dx$.

(d) Evaluate $\Gamma(-5/2)$.

(e) Define Power series and Taylor's series.

(f) Find the smallest positive period of the function $\sin(n\pi x/L)$.

(g) Are the following set of vectors [1, −1, 1], [1, 1, −1], [0, 1, 0] linearly independent or dependent?

(h) Define rank and nullity of a matrix.

(i) Find the spectral radius of the matrix $A=\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix}$.

(j) Show that the determinant of an orthogonal matrix has the value +1 or -1.

***Q.2 Solve both questions :*** **[14]**

(a) Verify Lagrange's mean value theorem for $f(x)=(x-1)(x-2)(x-3)$ in [0,4].

(b) Find the maxima and minima values of the function $f(x)=5x^6+18x^5+15x^4-10$.

***Q.3 Solve both questions :*** **[14]**

(a) Find the evolute of the parabola $y^2=4ax$.

(b) Evaluate the integral $\int_0^{\pi/2}\sqrt{(\tan\theta)}d\theta$.

***Q.4 Solve both questions :*** **[14]**

(a) Find the radius of convergence of the series $\sum_{m=0}^{\infty}(m+1)m\sim x^m$.

(b) Expand the function $f(x)=2x^3+7x^2+x-6$ in powers of $(x-2)$.

***Q.5 Solve both questions :*** **[14]**

(a) Find the half range sine Fourier series of the periodic function $f(x)=\pi-x, 0 < x <\pi$.

(b) Solve the system of equations by Cramer's rule:
$-x_1+x_2+2x_3=2$
$3x_1-x_2+x_3=6$
$-x_1+3x_2+4x_3=4$

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***Q.6 Solve both questions :*** **[14]**

(a) Find the inverse of the matrix $\begin{vmatrix} 3 & -1 & 5 \\ 2 & 6 & 4 \\ 5 & 5 & 9 \end{vmatrix}$ by Gauss-Jordan Elimination.

(b) Can we say all vectors in $R^3$ such that $4v_2+v_3=k$ form a vector space? If yes, determine the dimension and find a basis.

***Q.7 Solve both questions :*** **[14]**

(a) Find the nullity of the matrix $\begin{vmatrix} 4 & 0 & 2 & 8 \\ 5 & 7 & 3 & 1 \\ 0 & 6 & 9 & 0 \end{vmatrix}$.

(b) Find a basis of eigenvectors of the matrix A= $\begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix}$.

***Q.8 Solve both questions :*** **[14]**

(a) Find the eigenvalues and eigenvectors of the matrix A= $\begin{vmatrix} 5 & 4 \\ 1 & 2 \end{vmatrix}$.

(b) Diagonalize the matrix A= $\begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$.

***Q.9 Write short notes on any two of the following:*** **[14]**

(a) Mean value theorem

(b) Beta Function

(c) Vector Space

(d) Gauss Elimination

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